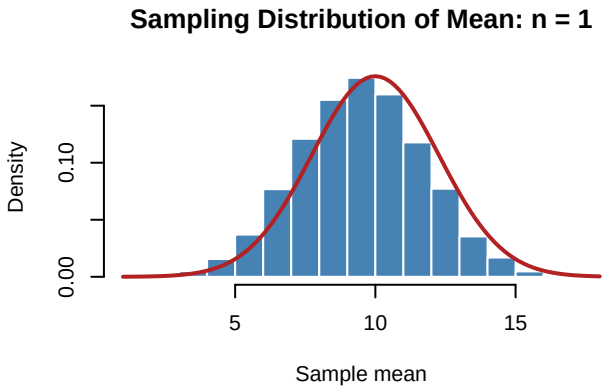
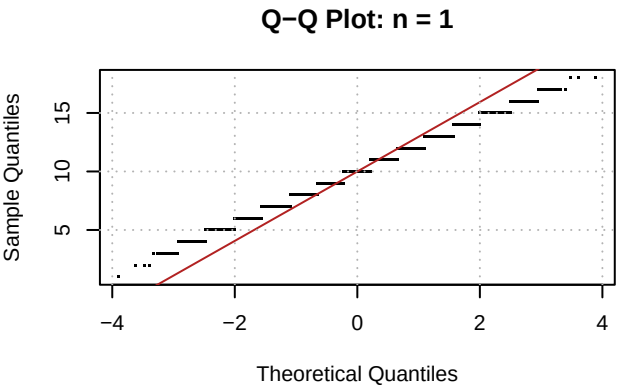
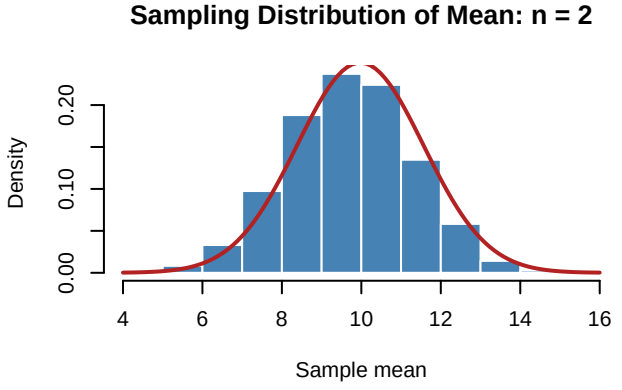
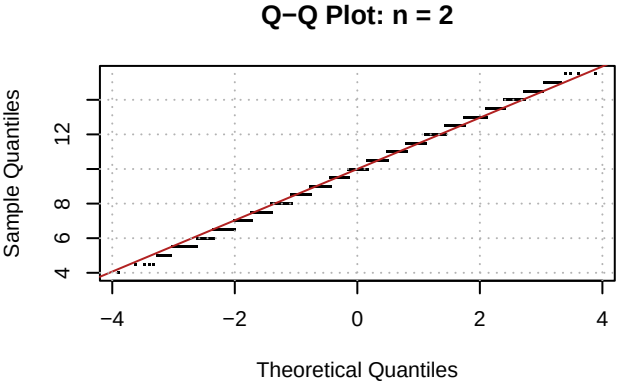
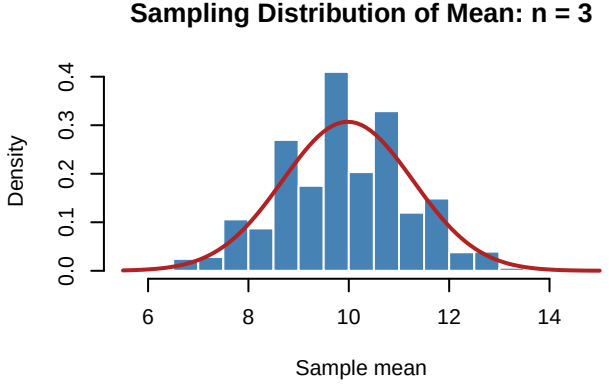
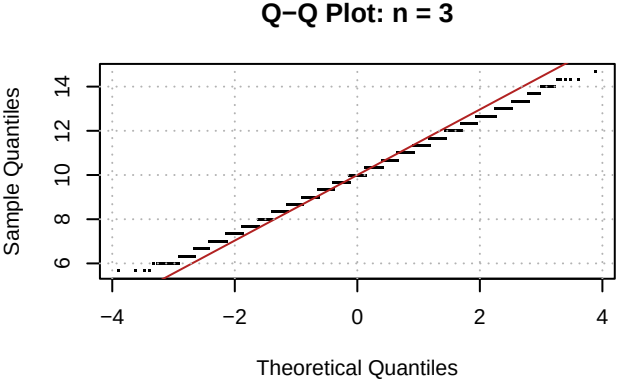

Sampling Distribution of Mean: n = 1
Density
0.00
0.10
5
10
15
Sample mean

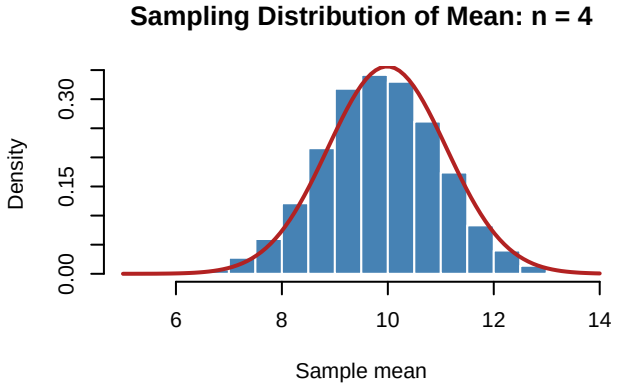

Q−Q Plot: n = 1
Sample Quantiles
5
10
15
−4
−2
0
2
4
Theoretical Quantiles

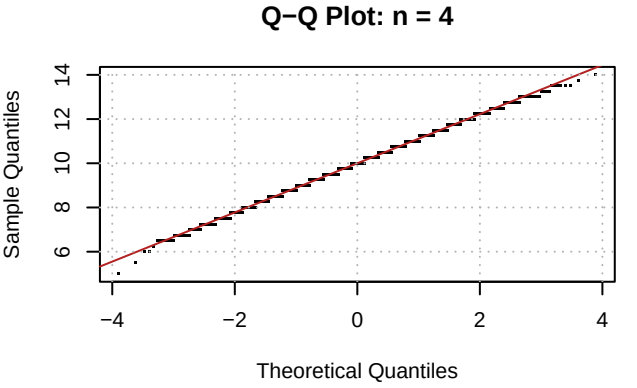

Sampling Distribution of Mean: n = 2
Density
0.00
0.10
0.20
4
6
8
10
12
14
16
Sample mean

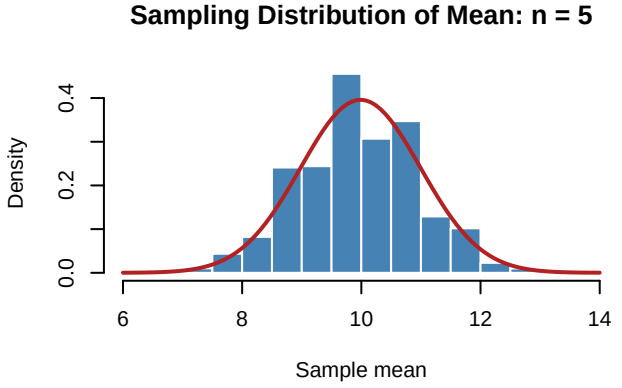

Q−Q Plot: n = 2
Sample Quantiles
4
6
8
12
−4
−2
0
2
4
Theoretical Quantiles

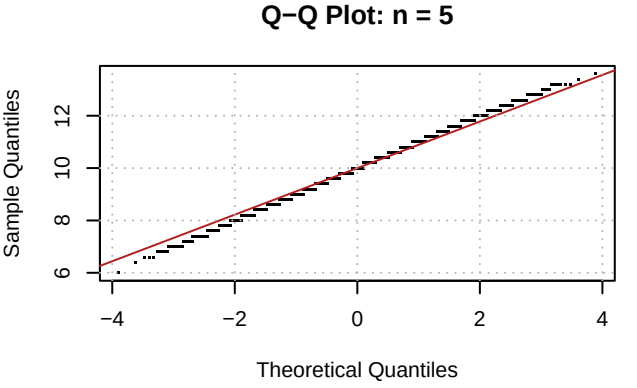

Sampling Distribution of Mean: n = 3
Density
0.0
0.1
0.2
0.3
0.4
6
8
10
12
14
Sample mean

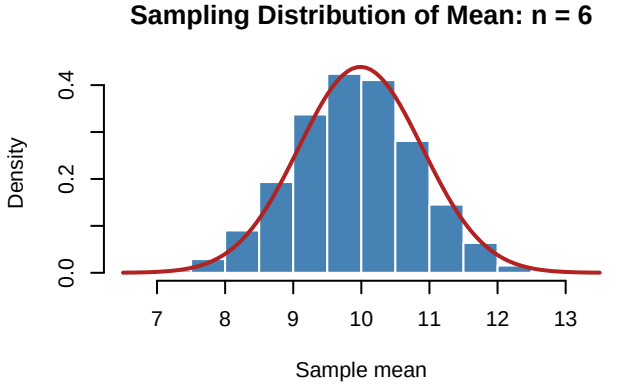

Q−Q Plot: n = 3
Sample Quantiles
6
8
10
12
14
−4
−2
0
2
4
Theoretical Quantiles

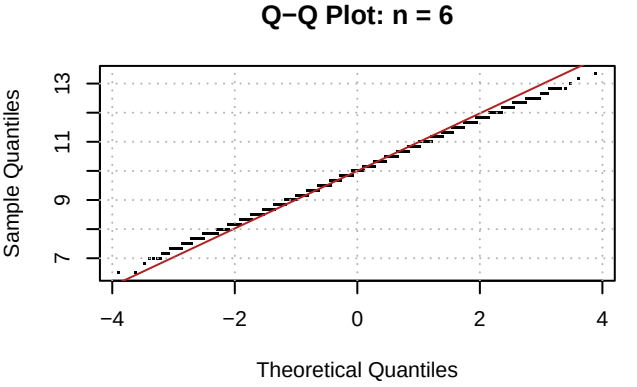

Sampling Distribution of Mean: n = 4
Density
0.00
0.15
0.30
6
8
10
12
14
Sample mean

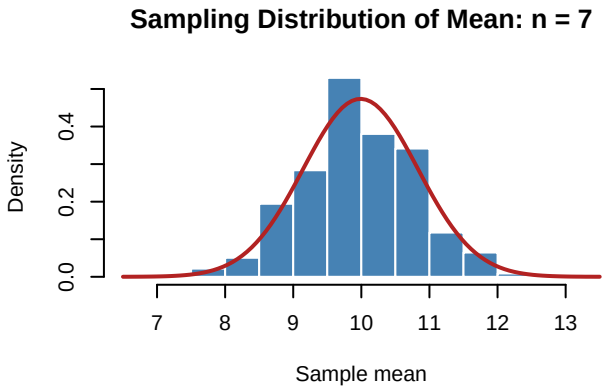

Q−Q Plot: n = 4
Sample Quantiles
6
8
10
12
14
−4
−2
0
2
4
Theoretical Quantiles

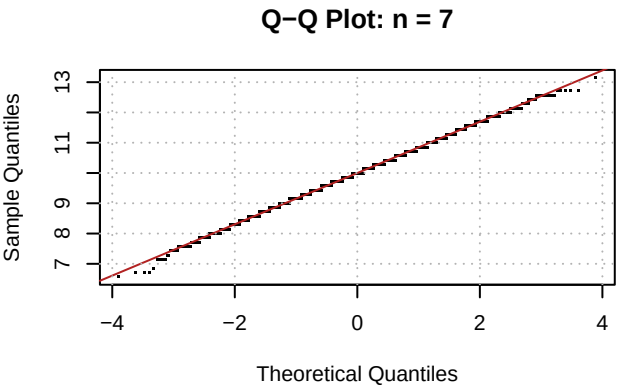

Sampling Distribution of Mean: n = 5
Density
0.0
0.2
0.4
6
8
10
12
14
Sample mean

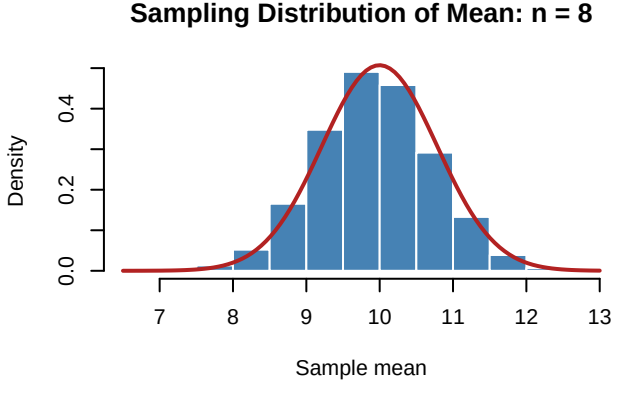

Q−Q Plot: n = 5
Sample Quantiles
6
8
10
12
−4
−2
0
2
4
Theoretical Quantiles

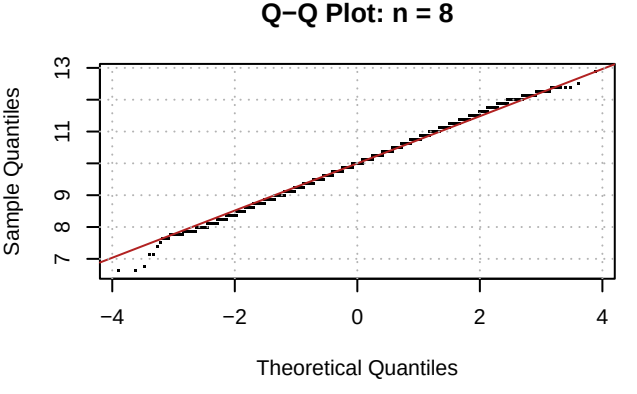

Sampling Distribution of Mean: n = 6
Density
0.0
0.2
0.4
7
8
9
10
11
12
13
Sample mean

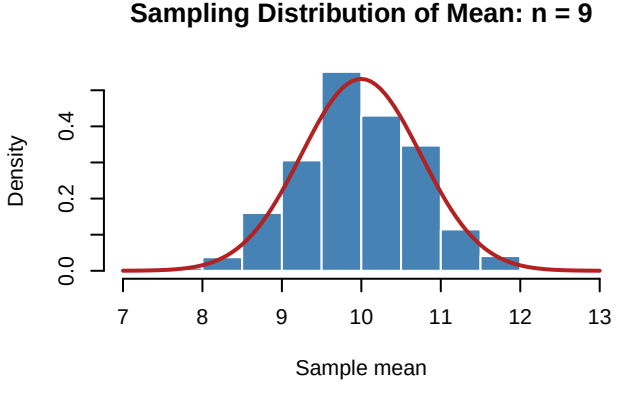

Q−Q Plot: n = 6
Sample Quantiles
7
9
11
13
−4
−2
0
2
4
Theoretical Quantiles

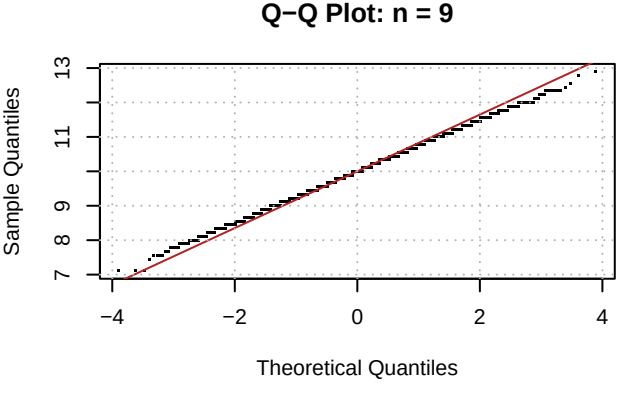

Sampling Distribution of Mean: n = 7
Density
0.0
0.2
0.4
7
8
9
10
11
12
13
Sample mean

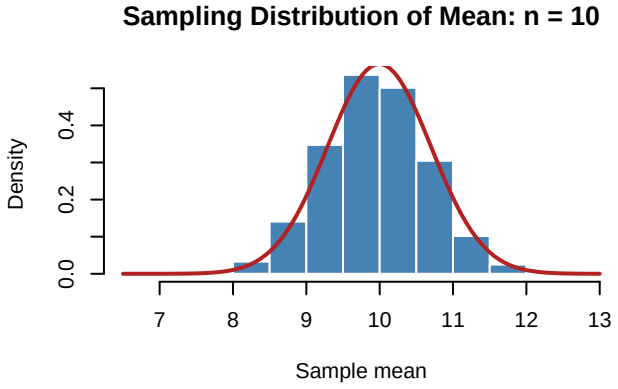

Q−Q Plot: n = 7
Sample Quantiles
7
8
9
11
13
−4
−2
0
2
4
Theoretical Quantiles

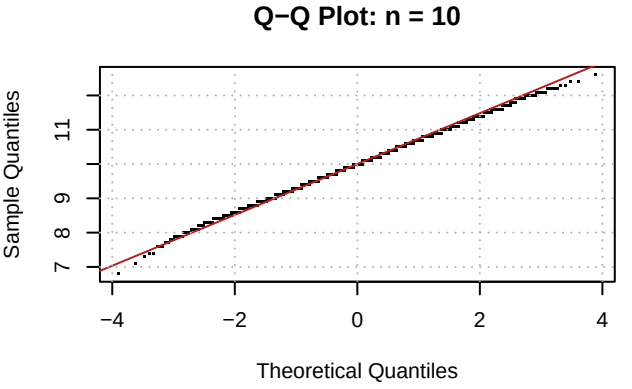

Sampling Distribution of Mean: n = 8
Density
0.0
0.2
0.4
7
8
9
10
11
12
13
Sample mean

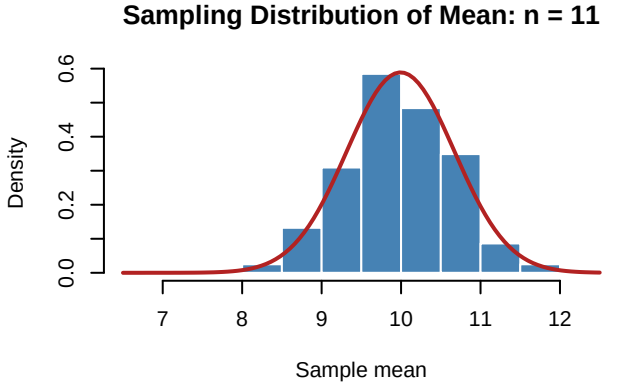

Q−Q Plot: n = 8
Sample Quantiles
7
8
9
11
13
−4
−2
0
2
4
Theoretical Quantiles

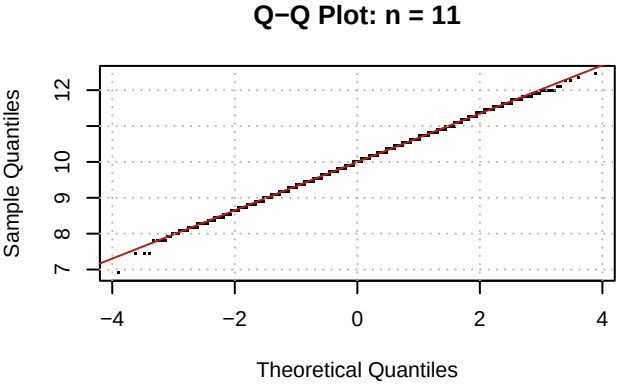

Sampling Distribution of Mean: n = 9
Density
0.0
0.2
0.4
7
8
9
10
11
12
13
Sample mean

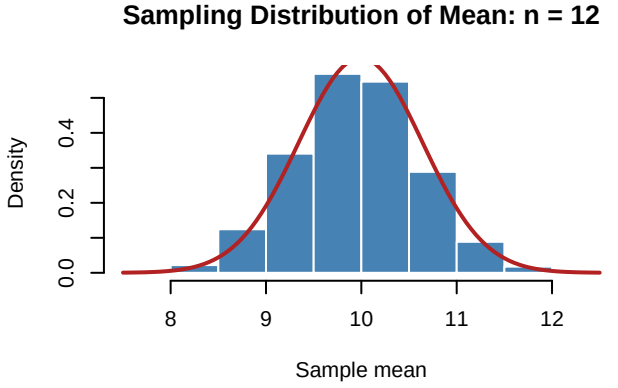

Q−Q Plot: n = 9
Sample Quantiles
7
8
9
11
13
−4
−2
0
2
4
Theoretical Quantiles

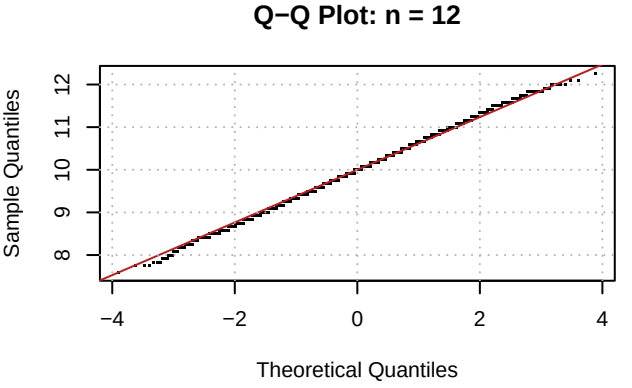

Sampling Distribution of Mean: n = 10
Density
0.0
0.2
0.4
7
8
9
10
11
12
13
Sample mean

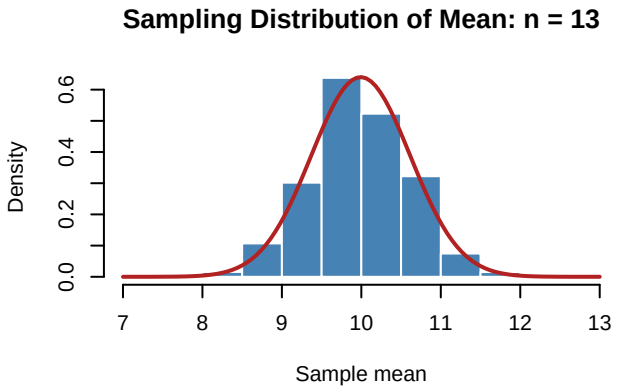

Q−Q Plot: n = 10
Sample Quantiles
7
8
9
11
−4
−2
0
2
4
Theoretical Quantiles

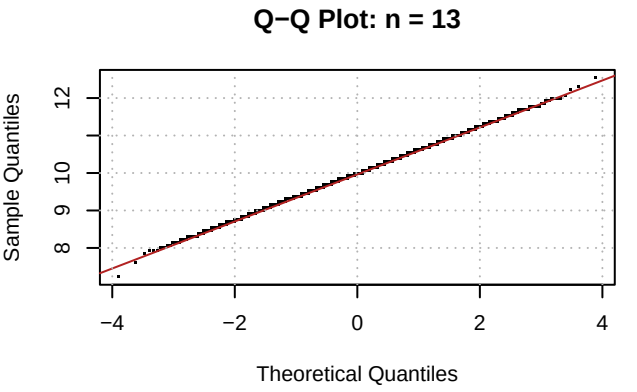

Sampling Distribution of Mean: n = 11
Density
0.0
0.2
0.4
0.6
7
8
9
10
11
12
Sample mean

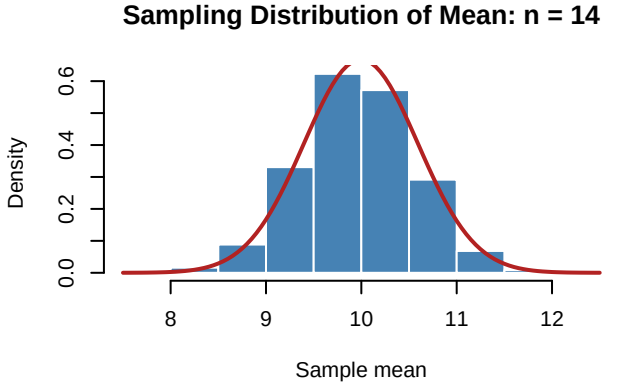

Q−Q Plot: n = 11
Sample Quantiles
7
8
9
10
12
−4
−2
0
2
4
Theoretical Quantiles

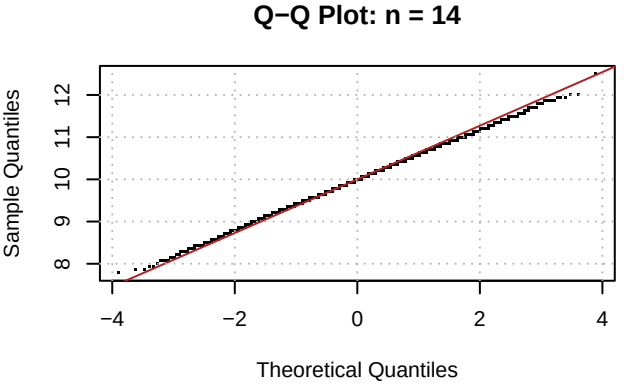

Sampling Distribution of Mean: n = 12
Density
0.0
0.2
0.4
8
9
10
11
12
Sample mean

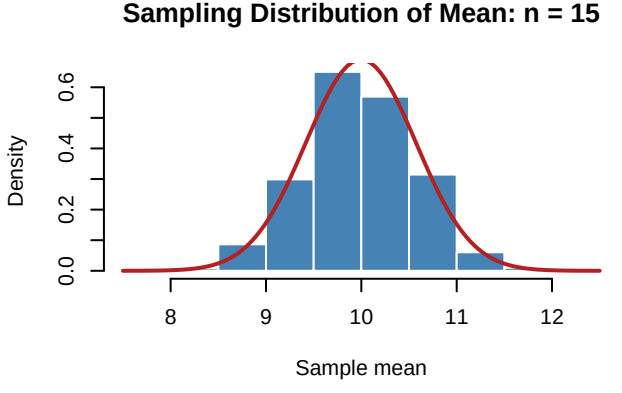

Q−Q Plot: n = 12
Sample Quantiles
8
9
10
11
12
−4
−2
0
2
4
Theoretical Quantiles

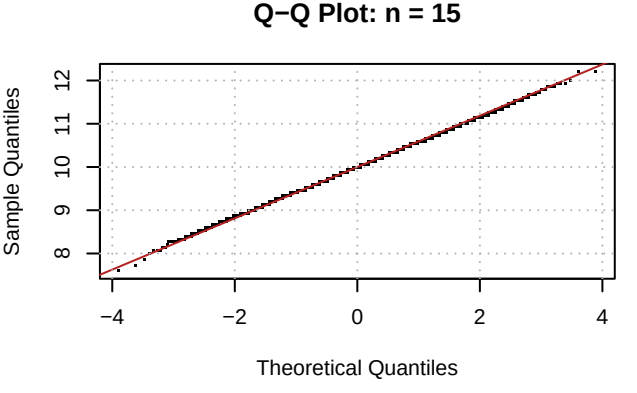

Sampling Distribution of Mean: n = 13
Density
0.0
0.2
0.4
0.6
7
8
9
10
11
12
13
Sample mean

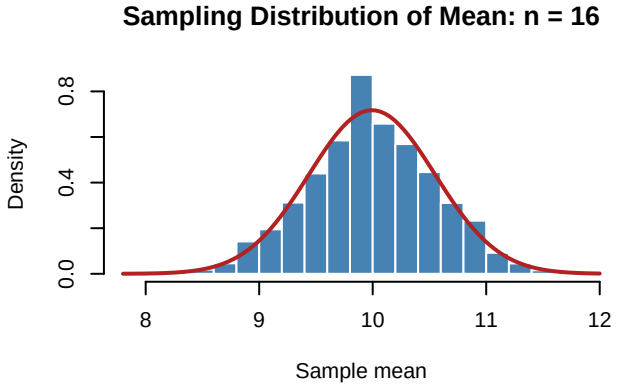

Q−Q Plot: n = 13
Sample Quantiles
8
9
10
12
−4
−2
0
2
4
Theoretical Quantiles

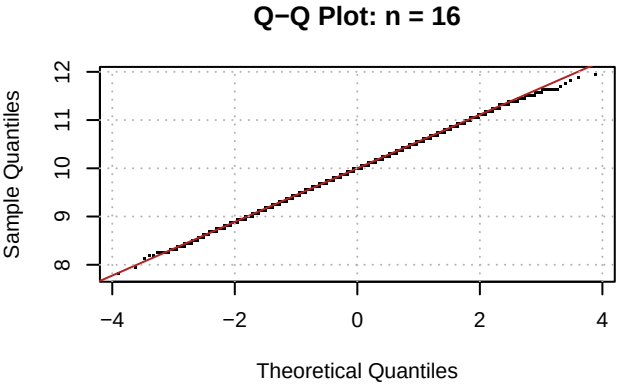

Sampling Distribution of Mean: n = 14
Density
0.0
0.2
0.4
0.6
8
9
10
11
12
Sample mean

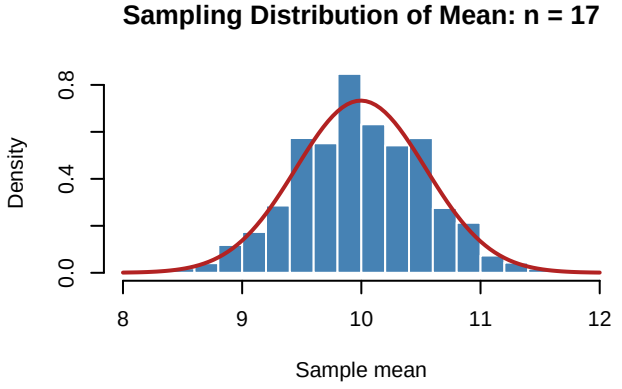

Q−Q Plot: n = 14
Sample Quantiles
8
9
10
11
12
−4
−2
0
2
4
Theoretical Quantiles

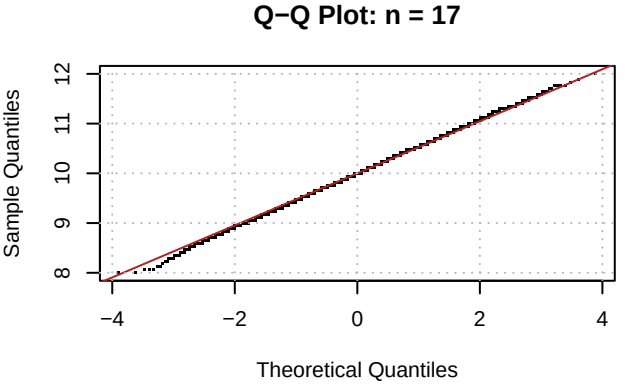

Sampling Distribution of Mean: n = 15
Density
0.0
0.2
0.4
0.6
8
9
10
11
12
Sample mean

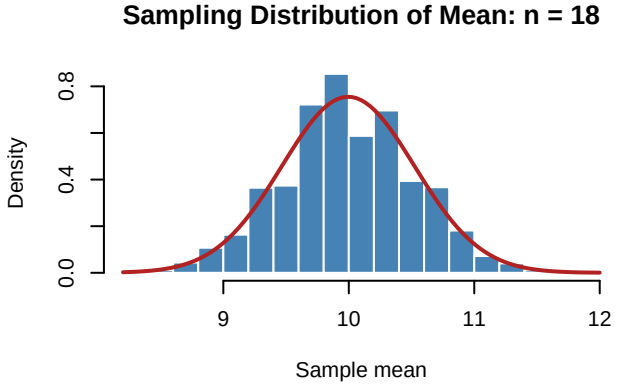

Q−Q Plot: n = 15
Sample Quantiles
8
9
10
11
12
−4
−2
0
2
4
Theoretical Quantiles

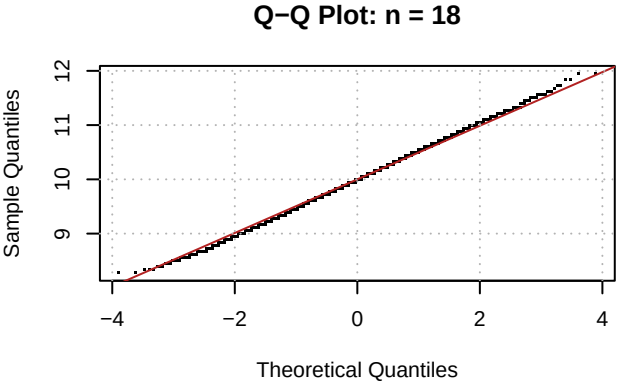

Sampling Distribution of Mean: n = 16
Density
0.0
0.4
0.8
8
9
10
11
12
Sample mean

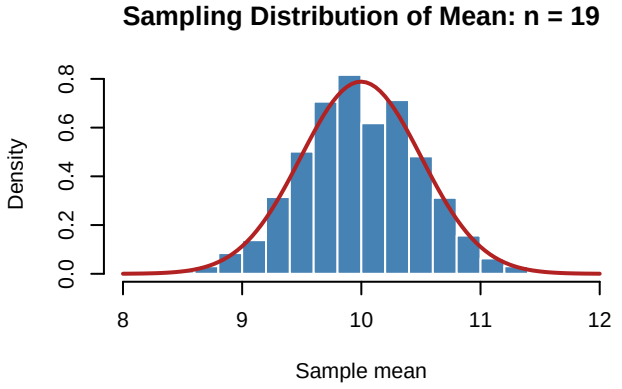

Q−Q Plot: n = 16
Sample Quantiles
8
9
10
11
12
−4
−2
0
2
4
Theoretical Quantiles

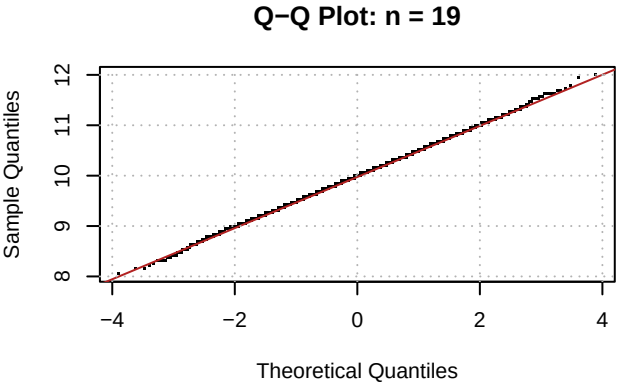

Sampling Distribution of Mean: n = 17
Density
0.0
0.4
0.8
8
9
10
11
12
Sample mean

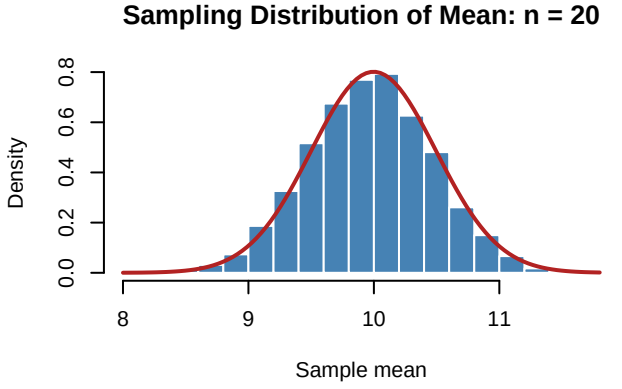

Q−Q Plot: n = 17
Sample Quantiles
8
9
10
11
12
−4
−2
0
2
4
Theoretical Quantiles

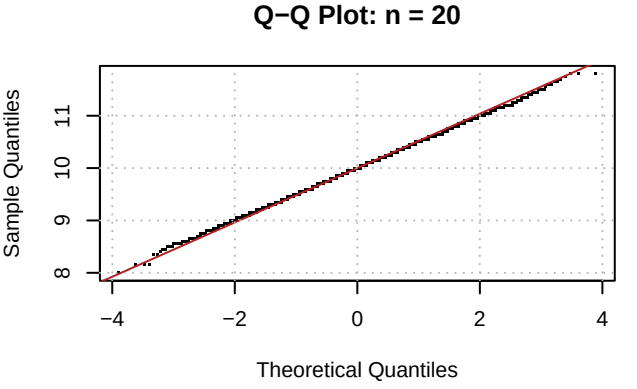

Sampling Distribution of Mean: n = 18
Density
0.0
0.4
0.8
9
10
11
12
Sample mean

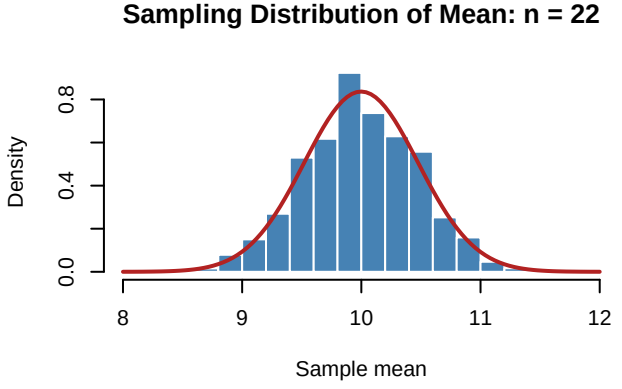

Q−Q Plot: n = 18
Sample Quantiles
9
10
11
12
−4
−2
0
2
4
Theoretical Quantiles

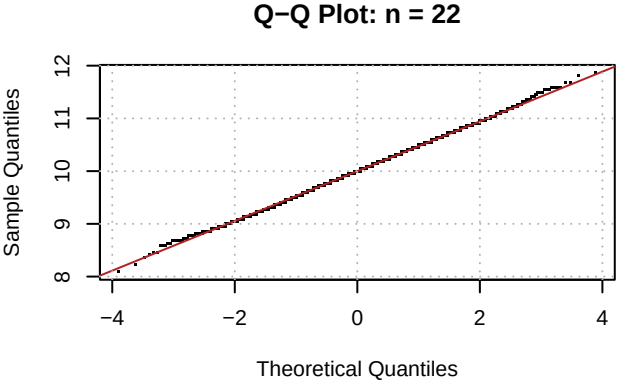

Sampling Distribution of Mean: n = 19
Density
0.0
0.2
0.4
0.6
0.8
8
9
10
11
12
Sample mean

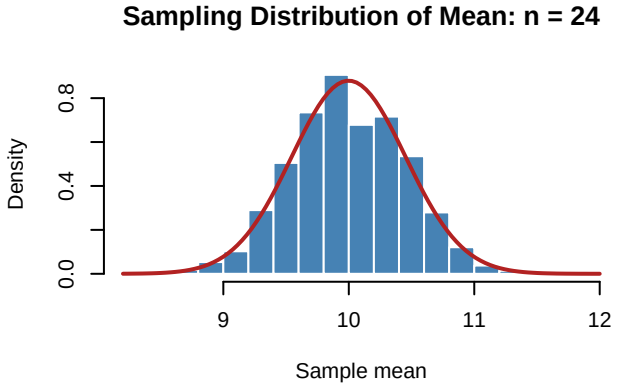

Q−Q Plot: n = 19
Sample Quantiles
8
9
10
11
12
−4
−2
0
2
4
Theoretical Quantiles

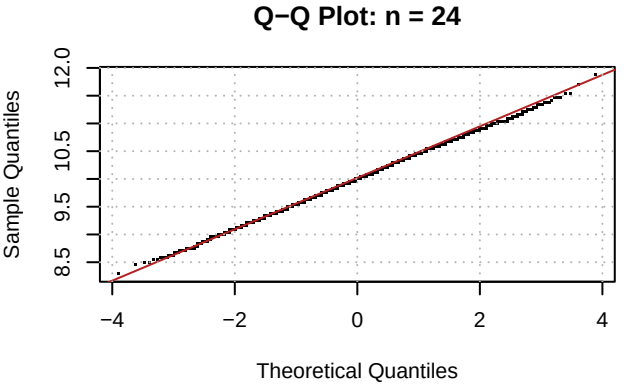

Sampling Distribution of Mean: n = 20
Density
0.0
0.2
0.4
0.6
0.8
8
9
10
11
Sample mean

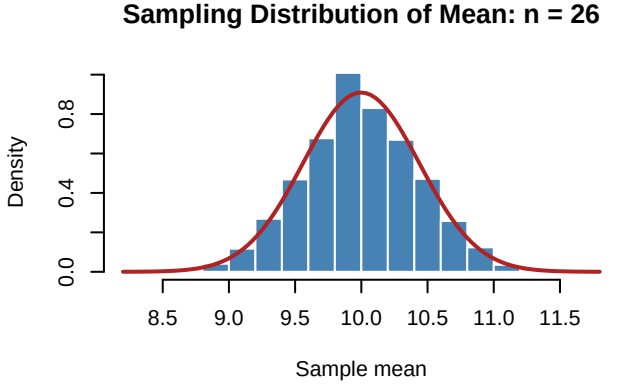

Q−Q Plot: n = 20
Sample Quantiles
8
9
10
11
−4
−2
0
2
4
Theoretical Quantiles

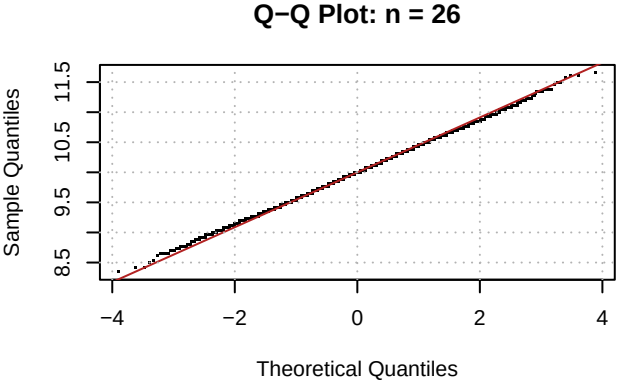

Sampling Distribution of Mean: n = 22
Density
0.0
0.4
0.8
8
9
10
11
12
Sample mean

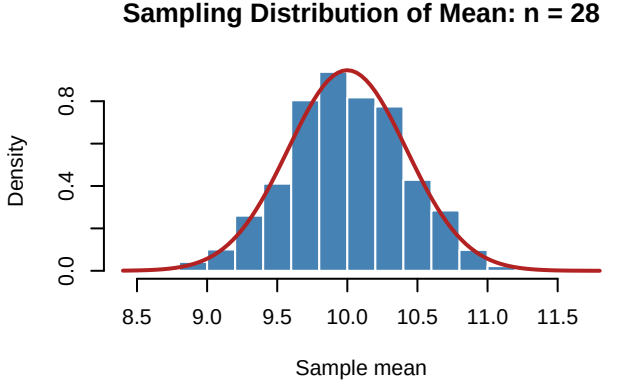

Q−Q Plot: n = 22
Sample Quantiles
8
9
10
11
12
−4
−2
0
2
4
Theoretical Quantiles

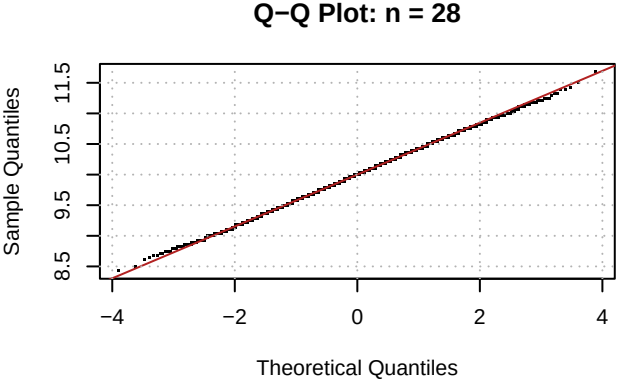

Sampling Distribution of Mean: n = 24
Density
0.0
0.4
0.8
9
10
11
12
Sample mean

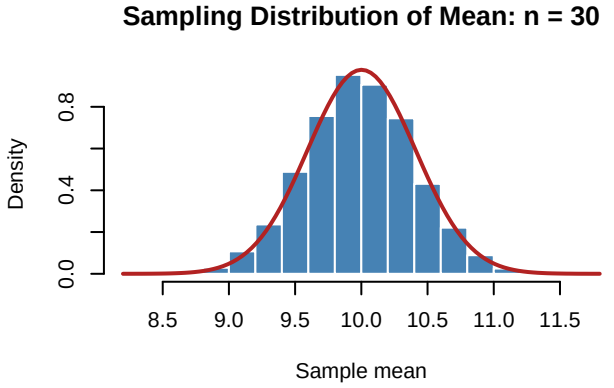

Q−Q Plot: n = 24
Sample Quantiles
8.5
9.5
10.5
12.0
−4
−2
0
2
4
Theoretical Quantiles

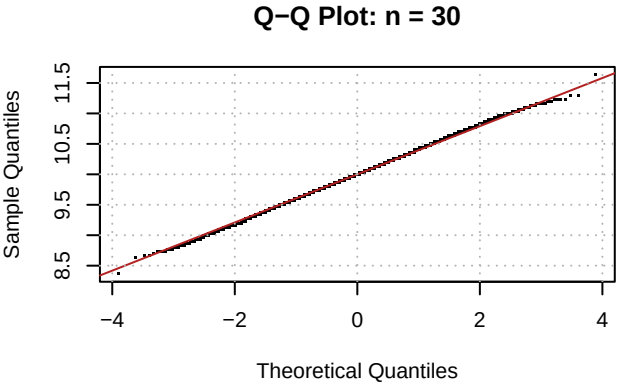

Sampling Distribution of Mean: n = 26
Density
0.0
0.4
0.8
8.5
9.0
9.5
10.0
10.5
11.0
11.5
Sample mean

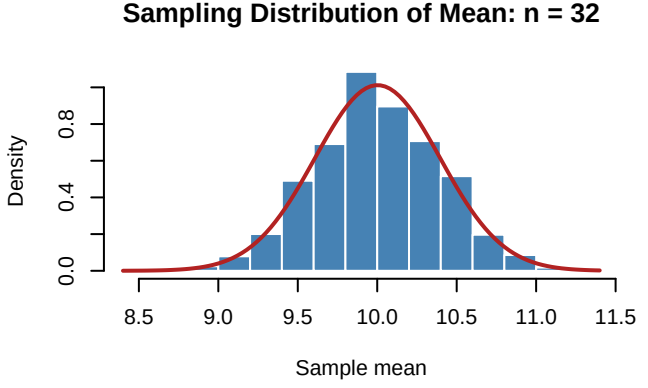

Q−Q Plot: n = 26
Sample Quantiles
8.5
9.5
10.5
11.5
−4
−2
0
2
4
Theoretical Quantiles

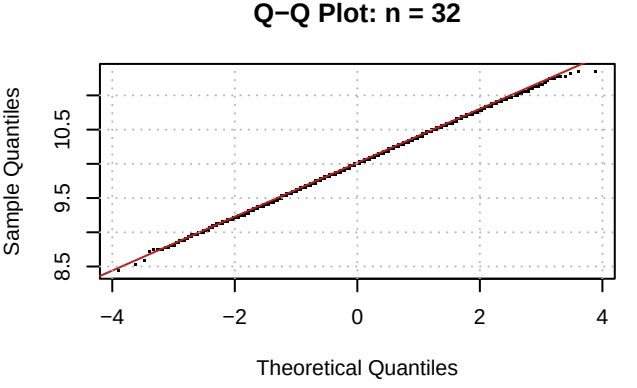

Sampling Distribution of Mean: n = 28
Density
0.0
0.4
0.8
8.5
9.0
9.5
10.0
10.5
11.0
11.5
Sample mean

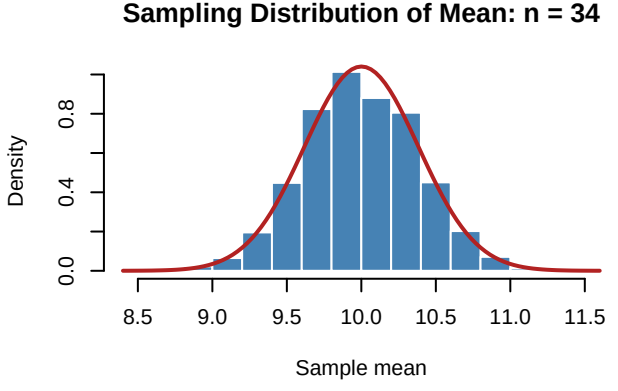

Q−Q Plot: n = 28
Sample Quantiles
8.5
9.5
10.5
11.5
−4
−2
0
2
4
Theoretical Quantiles

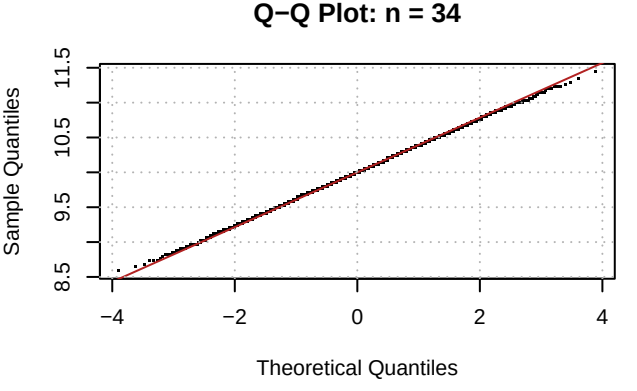

Sampling Distribution of Mean: n = 30
Density
0.0
0.4
0.8
8.5
9.0
9.5
10.0
10.5
11.0
11.5
Sample mean

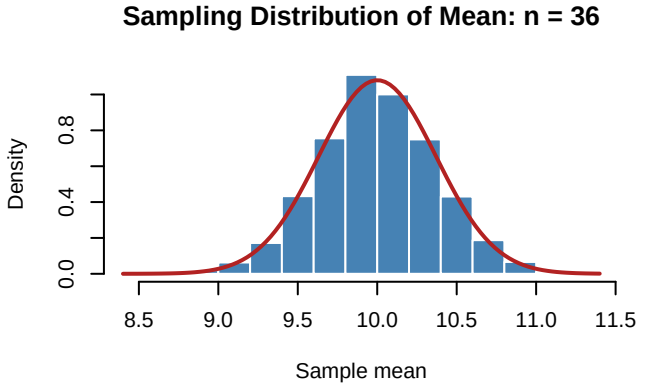

Q−Q Plot: n = 30
Sample Quantiles
8.5
9.5
10.5
11.5
−4
−2
0
2
4
Theoretical Quantiles

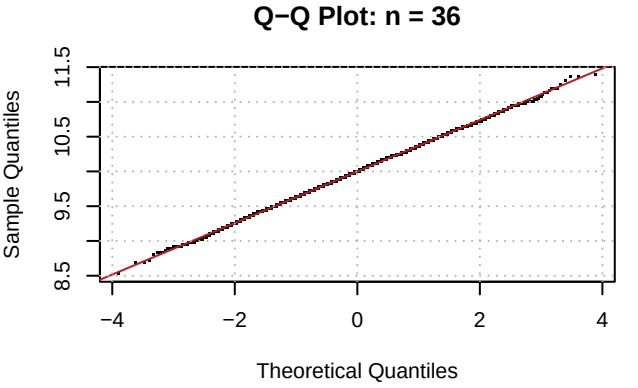

Sampling Distribution of Mean: n = 32
Density
0.0
0.4
0.8
8.5
9.0
9.5
10.0
10.5
11.0
11.5
Sample mean

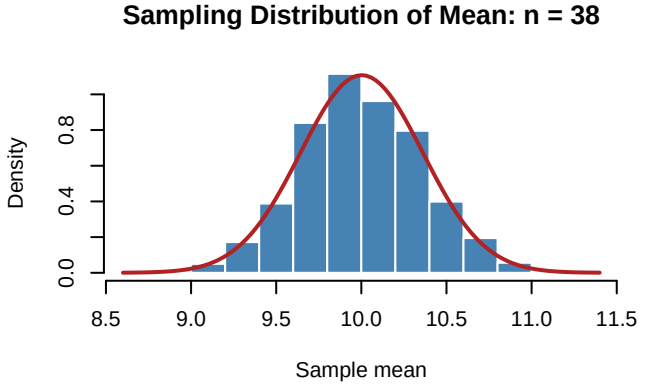

Q−Q Plot: n = 32
Sample Quantiles
8.5
9.5
10.5
−4
−2
0
2
4
Theoretical Quantiles

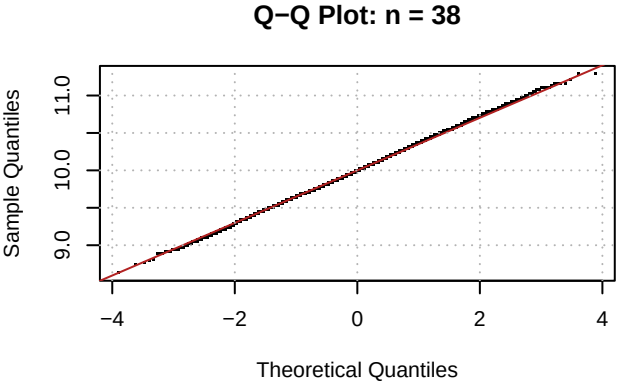

Sampling Distribution of Mean: n = 34
Density
0.0
0.4
0.8
8.5
9.0
9.5
10.0
10.5
11.0
11.5
Sample mean

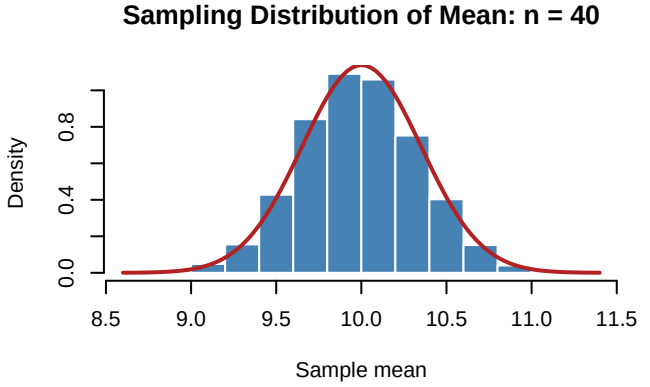

Q−Q Plot: n = 34
Sample Quantiles
8.5
9.5
10.5
11.5
−4
−2
0
2
4
Theoretical Quantiles

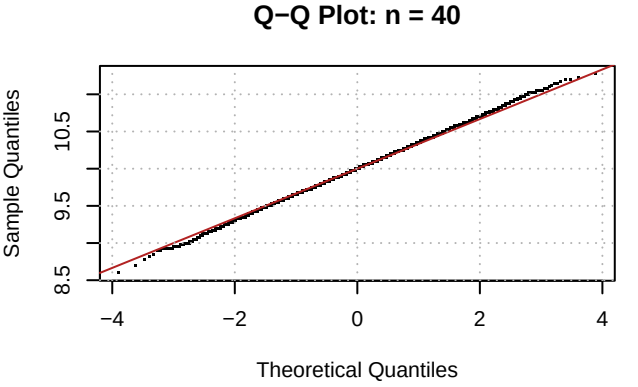

Sampling Distribution of Mean: n = 36
Density
0.0
0.4
0.8
8.5
9.0
9.5
10.0
10.5
11.0
11.5
Sample mean

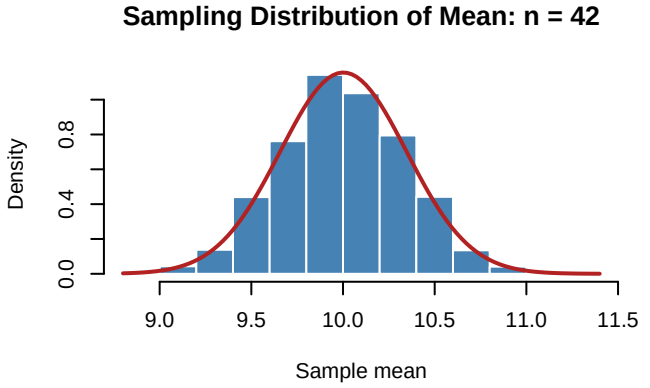

Q−Q Plot: n = 36
Sample Quantiles
8.5
9.5
10.5
11.5
−4
−2
0
2
4
Theoretical Quantiles

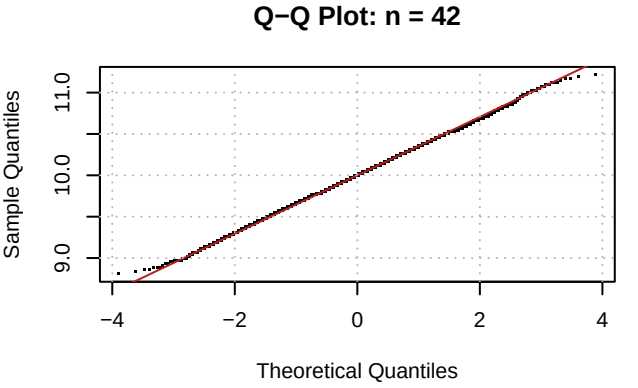

Sampling Distribution of Mean: n = 38
Density
0.0
0.4
0.8
8.5
9.0
9.5
10.0
10.5
11.0
11.5
Sample mean

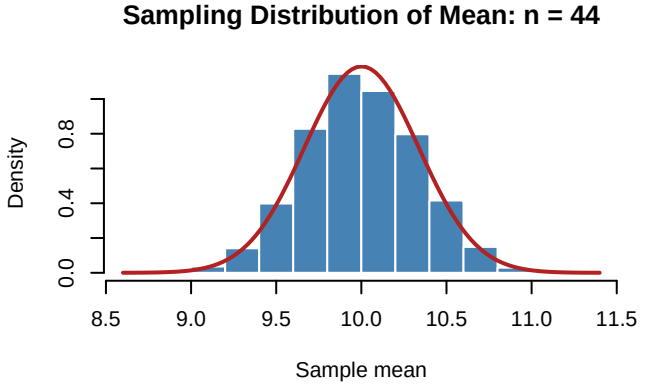

Q−Q Plot: n = 38
Sample Quantiles
9.0
10.0
11.0
−4
−2
0
2
4
Theoretical Quantiles

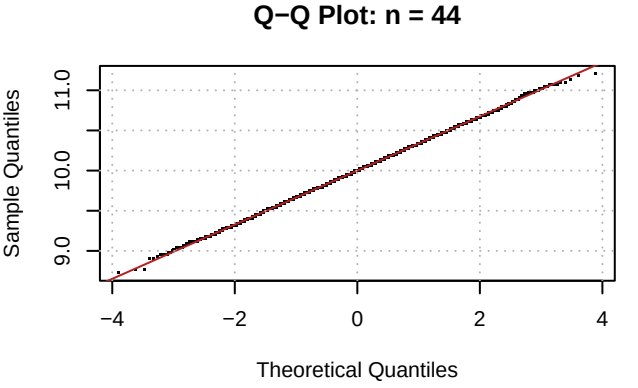

Sampling Distribution of Mean: n = 40
Density
0.0
0.4
0.8
8.5
9.0
9.5
10.0
10.5
11.0
11.5
Sample mean

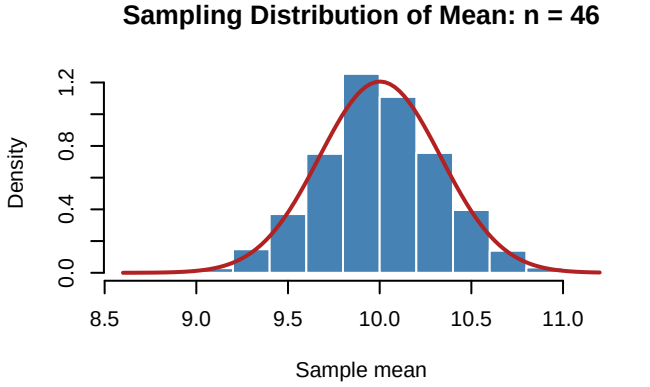

Q−Q Plot: n = 40
Sample Quantiles
8.5
9.5
10.5
−4
−2
0
2
4
Theoretical Quantiles

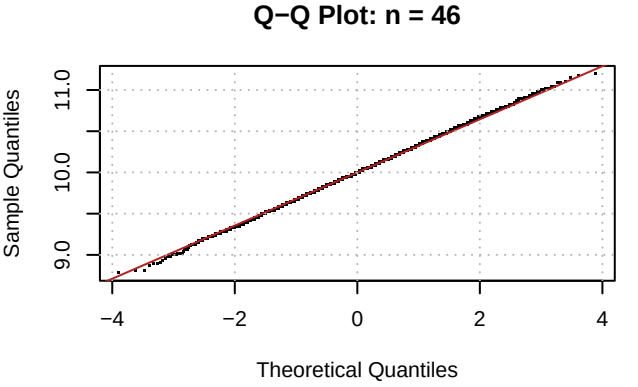

Sampling Distribution of Mean: n = 42
Density
0.0
0.4
0.8
9.0
9.5
10.0
10.5
11.0
11.5
Sample mean

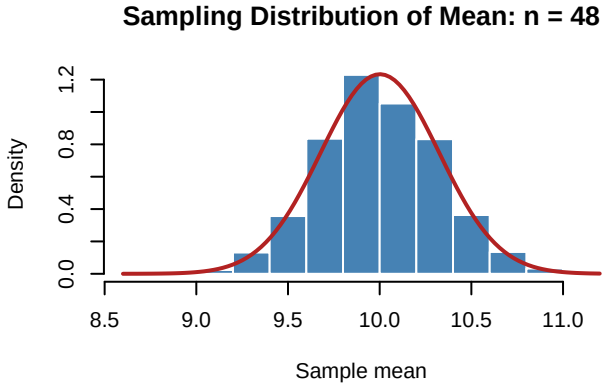

Q−Q Plot: n = 42
Sample Quantiles
9.0
10.0
11.0
−4
−2
0
2
4
Theoretical Quantiles

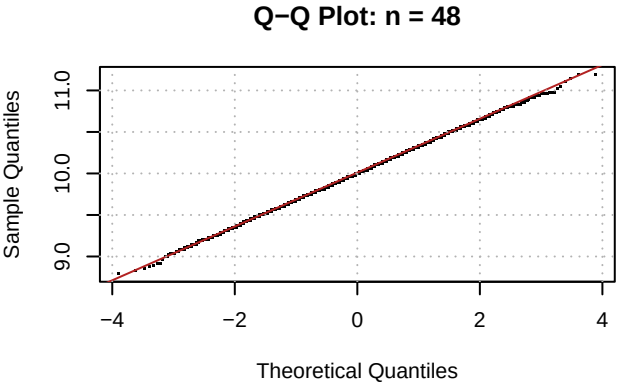

Sampling Distribution of Mean: n = 44
Density
0.0
0.4
0.8
8.5
9.0
9.5
10.0
10.5
11.0
11.5
Sample mean

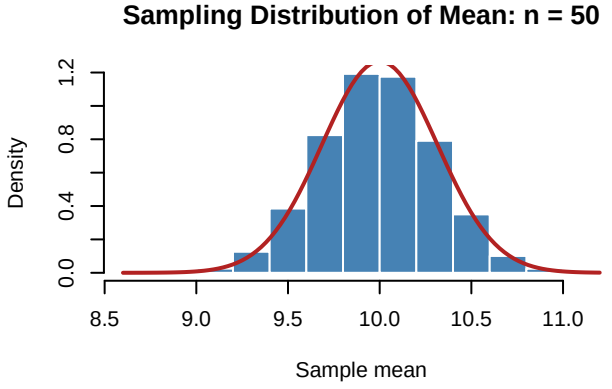

Q−Q Plot: n = 44
Sample Quantiles
9.0
10.0
11.0
−4
−2
0
2
4
Theoretical Quantiles

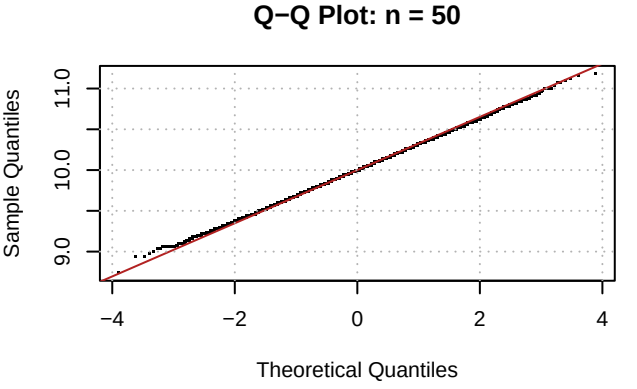

Sampling Distribution of Mean: n = 46
Density
0.0
0.4
0.8
1.2
8.5
9.0
9.5
10.0
10.5
11.0
Sample mean

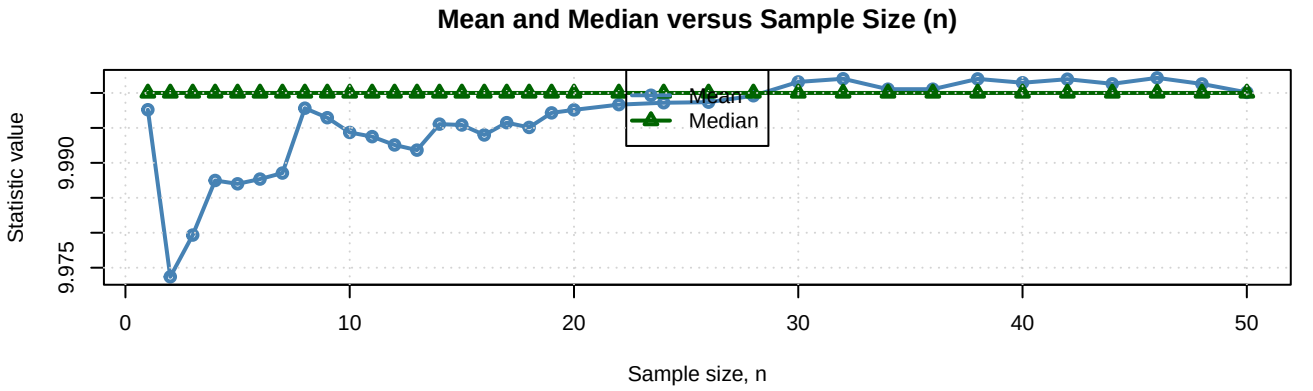

Q−Q Plot: n = 46
Sample Quantiles
9.0
10.0
11.0
−4
−2
0
2
4
Theoretical Quantiles

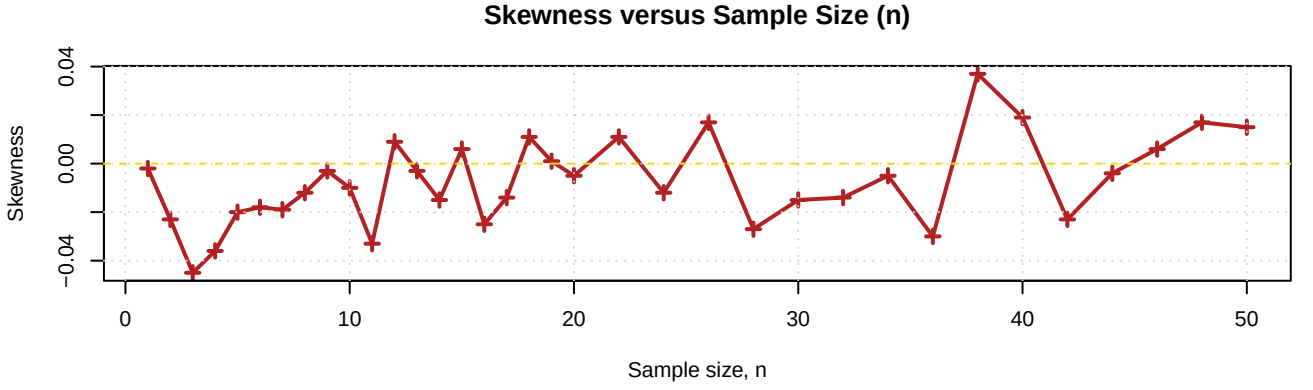

Sampling Distribution of Mean: n = 48
Density
0.0
0.4
0.8
1.2
8.5
9.0
9.5
10.0
10.5
11.0
Sample mean

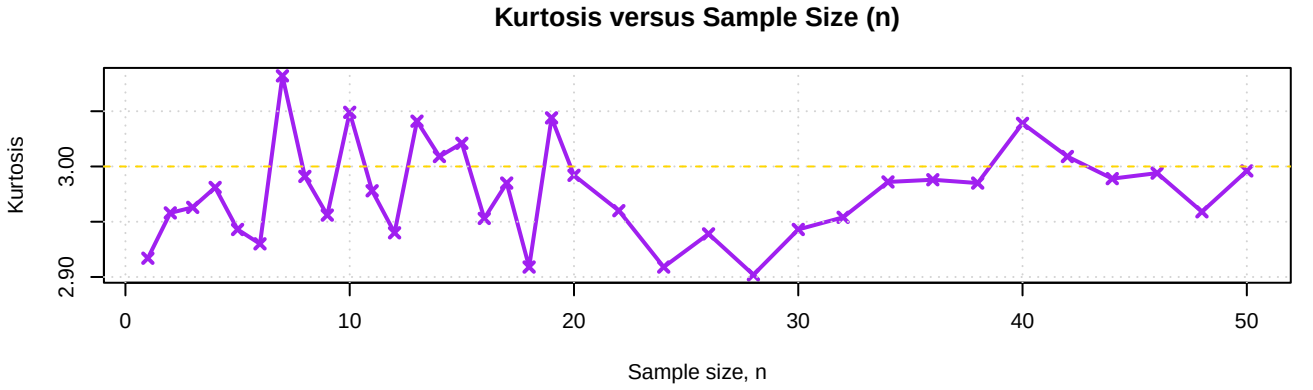

Q−Q Plot: n = 48
Sample Quantiles
9.0
10.0
11.0
−4
−2
0
2
4
Theoretical Quantiles


Sampling Distribution of Mean: n = 50
Density
0.0
0.4
0.8
1.2
8.5
9.0
9.5
10.0
10.5
11.0
Sample mean


Q−Q Plot: n = 50
Sample Quantiles
9.0
10.0
11.0
−4
−2
0
2
4
Theoretical Quantiles


Mean and Median versus Sample Size (n)
Statistic value
9.975
9.990
Mean
Median
0
10
20
30
40
50
Sample size, n


Skewness versus Sample Size (n)
Skewness
−0.04
0.00
0.04
0
10
20
30
40
50
Sample size, n


Kurtosis versus Sample Size (n)
Kurtosis
2.90
3.00
0
10
20
30
40
50
Sample size, n